APPENDIX E

Theorems Concerning Product Wave Functions

In this appendix we shall derive some theorems and formulae used in connection with the general single-particle model (Chap. 7) and in more extended theories that are based on it (Chap. 10).

E.1 The Bloch–Messiah Theorem [BM 62]

The Bloch–Messiah theorem states that:

A unitary matrix $\mathcal{W}$ of the following special form can be decomposed into three matrices:

$$\mathcal{W} = \begin{pmatrix} U & V^* \\ V & U^* \end{pmatrix} = \begin{pmatrix} D & 0 \\ 0 & D^* \end{pmatrix} \begin{pmatrix} \overline{U} & \overline{V} \\ \overline{V} & \overline{U} \end{pmatrix} \begin{pmatrix} C & 0 \\ 0 & C^* \end{pmatrix}, \tag{E.1}$$

where the real matrices $\overline{U}$ and $\overline{V}$ are diagonal and of "quasi-canonical" form, as given in Eq. (7.9).

The proof is based on the property that an Hermitian matrix ρ can be diagonalized by a unitary transformation D such that

$$\bar{\rho} = D^+ \rho D \quad \text{is diagonal with real eigenvalues } \rho_k, \tag{E.2}$$

and that a skew symmetric matrix κ can be brought into canonical form by (generally different) unitary matrix D:

$$\bar{\kappa} = D^+ \kappa D^* \quad \text{decomposes into } (2\times 2) \text{ boxes of the form } \begin{pmatrix} 0 & \kappa_k \\ -\kappa_k & 0 \end{pmatrix} \tag{E.3}$$

with real numbers κ_k, along the diagonal line [Zu 62]. In the next step, we show that if we have the relation

$$\rho\kappa = \kappa\rho^* \tag{E.4}$$

we can find *one* matrix D which diagonalizes ρ and brings κ into canonical form. For this purpose we first diagonalize ρ and from (E.4) obtain in this basis

$$(\rho_i - \rho_k)\kappa_{ik} = 0. \tag{E.5}$$

This means κ_{ik} vanishes in this basis for all values i and k with $\rho_i \neq \rho_k$, that is, we can restrict ourselves to subspaces of degenerate eigenvalues ρ_i. The κ does not mix them. In these subspaces ρ is a multiple of the unity. We can therefore bring κ into canonical form in each subspace without changing the diagonal character of ρ.

We now apply these general considerations to the transformation (E.1). First we suppose that U is Hermitian and positive semi-definite, otherwise this can be achieved by a suitable C-transformation, for instance,

$$C = (\sqrt{UU^+})^{-1} U^+. \tag{E.6}$$

From Eq. (7.5) we see that the matrices

$$\rho = V^* V^T = 1 - UU^+ \tag{E.7}$$

and

$$\kappa = V^* U^T = -\kappa^T \tag{E.8}$$

fulfill the condition (E.4). Therefore, we can find a basis (given by the transformation D) in which κ is in canonical form and ρ as well as U are diagonal with real eigenvalues v_k^2 and $u_k \geqslant 0$. From (E.7) we get the normalization

$$u_k^2 + v_k^2 = 1. \tag{E.9}$$

We now have to distinguish between eigenspaces of U with eigenvalues $u_k \neq 0$ and those with eigenvalue 0. In the first case we can divide κ_{ik} by u_k and find from Eq. (E.8) that V is in canonical form with elements v_k. In the second case we see from Eq. (E.7) that V is unitary in these subspaces and we can shift it either to the matrix D or to the matrix C. The rest is a unity matrix.

In particular, we see from this derivation that the numbers u_k and v_k are real. Any complex phases can be shifted to the transformations C and D. As long as we restrict ourselves to real coefficients D, however, it may be interesting to use relative phases between u_k and v_k. The corresponding BCS wave functions will then depend on them.

E.2 Operators in the Quasi-particle Space

Using the inverse transformation to (7.1) we can transform all operators expressed by c^+, c in the quasi-particle space. We now give a few examples:

For a Hermitian one-particle operator

$$\hat{F}=\sum_{ll'} f_{ll'}c_l^+c_{l'}+\tfrac{1}{2}(g_{ll'}c_l^+c_{l'}^++\text{h.c.}) \tag{E.10}$$

we get

$$\hat{F}=F^0+\sum_{kk'}F_{kk'}^{11}\beta_k^+\beta_{k'}+\tfrac{1}{2}\sum_{kk'}(F_{kk'}^{20}\beta_k^+\beta_{k'}^++\text{h.c.}) \tag{E.11}$$

with

$$F^0=\mathrm{Tr}(f\rho)-\tfrac{1}{2}\mathrm{Tr}(g\kappa^*+g^*\kappa), \tag{E.12}$$

$$F_{kk'}^{11}=(U^+fU-\ V^+f^TV+\ U^+gV-\ V^+g^*U)_{kk'}, \tag{E.13}$$

$$F_{kk'}^{20}=(U^+fV^*-V^+f^TU^*+U^+gU^*-V^+g^*V^*)_{kk'}. \tag{E.14}$$

In the case of time-reversal invariance $T\hat{F}T^+=\tau\hat{F}$ with $\tau=\pm1$ this means in the canonical basis (see Sec. 7.2.1)

$$F_{kk'}^{11}=\ f_{kk'}\xi_{kk'}^{\tau}-g_{k\bar{k}'}\eta_{kk'}^{\tau} \tag{E.15}$$

$$F_{kk'}^{20}=-f_{k\bar{k}'}\eta_{kk'}^{\tau}+g_{kk'}\xi_{kk'}^{\tau} \tag{E.16}$$

where $\xi^{\pm}$ and $\eta^{\pm}$ are defined in Eq. (8.202).

An example of a Hermitian two-particle operator is the Hamiltonian

$$\hat{H}=\sum_{l_1l_2}\epsilon_{l_1l_2}c_{l_1}^+c_{l_2}+\tfrac{1}{4}\sum_{l_1l_2l_3l_4}\bar{v}_{l_1l_2l_3l_4}c_{l_1}^+c_{l_2}^+c_{l_4}c_{l_3}. \tag{E.17}$$

We get

$$\begin{aligned}\hat{H}=H^0&+\sum_{k_1k_2}H_{k_1k_2}^{11}\beta_{k_1}^+\beta_{k_2}+\tfrac{1}{2}\sum_{k_1k_2}(H_{k_1k_2}^{20}\beta_{k_1}^+\beta_{k_2}+\text{h.c.})\\&+\sum_{k_1k_2k_3k_4}(H_{k_1k_2k_3k_4}^{40}\beta_{k_1}^+\beta_{k_2}^+\beta_{k_3}^+\beta_{k_4}^++\text{h.c.})\\&+\sum_{k_1k_2k_3k_4}(H_{k_1k_2k_3k_4}^{31}\beta_{k_1}^+\beta_{k_2}^+\beta_{k_3}^+\beta_{k_4}+\text{h.c.})\\&+\tfrac{1}{4}\sum_{k_1k_2k_3k_4}H_{k_1k_2k_3k_4}^{22}\beta_{k_1}^+\beta_{k_2}^+\beta_{k_4}\beta_{k_3}.\end{aligned} \tag{E.18}$$

With the definitions

$$h=\epsilon+\Gamma, \tag{E.19a}$$

$$\Gamma_{lm}=\sum_{pq}\bar{v}_{lqmp}\rho_{pq}\quad:=\mathrm{Tr}_1(\bar{v}\rho), \tag{E.19b}$$

$$\Delta_{lm}=\tfrac{1}{2}\sum_{pq}\bar{v}_{lmpq}\kappa_{pq}:=-\tfrac{1}{2}\mathrm{Tr}_2(\bar{v}\kappa), \tag{E.19c}$$

we can write:

$$H^0 = \mathrm{Tr}(\epsilon\rho + \tfrac{1}{2}\Gamma\rho - \tfrac{1}{2}\Delta\kappa^*)$$
$$= \mathrm{Tr}(\epsilon\rho) + \tfrac{1}{2}\mathrm{Tr}_1\mathrm{Tr}_1(\rho\bar{v}\rho) + \tfrac{1}{4}\mathrm{Tr}_2\mathrm{Tr}_2(\kappa^*\bar{v}\kappa), \tag{E.20}$$

$$H^{11} = U^+hU - V^+h^TV + U^+\Delta V^* - V^+\Delta^*U, \tag{E.21}$$

$$H^{20} = U^+hV^* - V^+h^TU^* + U^+\Delta U^* - V^+\Delta^*V^*, \tag{E.22}$$

and

$$H^{40}_{k_1k_2k_3k_4} = \tfrac{1}{4}\sum_{l_1l_2l_3l_4}\bar{v}_{l_1l_2l_3l_4}U^*_{l_1k_1}U^*_{l_2k_2}V^*_{l_4k_3}V^*_{l_3k_4}, \tag{E.23a}$$

$$H^{31}_{k_1k_2k_3k_4} = \tfrac{1}{2}\sum_{l_1l_2l_3l_4}\bar{v}_{l_1l_2l_3l_4}(U^*_{l_1k_1}V^*_{l_4k_2}V^*_{l_3k_3}V_{l_2k_4} + V^*_{l_3k_1}U^*_{l_2k_2}U^*_{l_1k_3}U_{l_4k_4}), \tag{E.23b}$$

$$H^{22}_{k_1k_2k_3k_4} = \sum_{l_1l_2l_3l_4}\bar{v}_{l_1l_2l_3l_4}((U^*_{l_1k_1}V^*_{l_4k_2}V_{l_2k_3}U_{l_3k_4} - (k_1 \leftrightarrow k_2)) - (k_3 \leftrightarrow k_4)$$
$$+ U^*_{l_1k_1}U^*_{l_2k_2}U_{l_3k_3}U_{l_4k_4} + V^*_{l_3k_1}V^*_{l_4k_2}V_{l_1k_3}V_{l_2k_4}), \tag{E.23c}$$

Representations of these matrix elements in the canonical basis are given in Eqs. (8.201) and (9.133).

In the case of vanishing pairing correlations, the quasi-particles are either particles or holes:

$$\beta_m^+ = a_m^+ \quad \text{for } \epsilon_m > \epsilon_F; \qquad \beta_i^+ = a_i \qquad \text{for } \epsilon_i < \epsilon_F.$$

In this representation, the Hamiltonian (E.18) has the form:

$$H^0 = \sum_i\left(\epsilon_{ii} + \tfrac{1}{2}\sum_j\bar{v}_{ijij}\right), \tag{E.24a}$$

$$H^{11} = \sum_{mm'}\left(\epsilon_{mm'} + \sum_j\bar{v}_{mjm'j}\right)\beta_m^+\beta_{m'} - \sum_{ii'}\left(\epsilon_{i'i} + \sum_j\bar{v}_{i'jij}\right)\beta_i^+\beta_{i'}, \tag{E.24b}$$

$$H^{20} = \sum_{mi}\left(\epsilon_{mi} + \sum_j\bar{v}_{mjij}\right)\beta_m^+\beta_i^+ + \text{h.c.}, \tag{E.24c}$$

$$H^{40} = \tfrac{1}{4}\sum_{\substack{mn\\ij}}\bar{v}_{mnij}\beta_m^+\beta_i^+\beta_n^+\beta_j^+ + \text{h.c.}, \tag{E.25a}$$

$$H^{31} = \tfrac{1}{2}\sum_{mi}\beta_m^+\beta_i^+\left(\sum_{nn'}\bar{v}_{mnin'}\beta_n^+\beta_{n'} - \sum_{jj'}\bar{v}_{mj'ij}\beta_j^+\beta_{j'}\right) + \text{h.c.}, \tag{E.25b}$$

$$H^{22} = \tfrac{1}{4}\sum_{mm'nn'}\bar{v}_{mnm'n'}\beta_m^+\beta_n^+\beta_{n'}\beta_{m'} + \tfrac{1}{4}\sum_{iji'j'}\bar{v}_{iji'j'}\beta_{i'}^+\beta_{j'}^+\beta_j\beta_i$$
$$+ \sum_{minj}\bar{v}_{mjin}\beta_m^+\beta_i^+\beta_j\beta_n. \tag{E.25c}$$

E.3 Thouless' Theorem

Thouless' theorem [Th 60] states that:

Starting with a general product wave function $|\Phi_0\rangle$ *which is the vacuum to quasi-particle operators* β, *any other general product wave function* $|\Phi_1\rangle$ *which is not orthogonal to* $|\Phi_0\rangle$ *may be expressed in the form*

$$|\Phi_1\rangle = \mathfrak{N} \cdot \exp\left\{ \sum_{k<k'} Z_{kk'} \beta_k^+ \beta_{k'}^+ \right\} |\Phi_0\rangle, \tag{E.26}$$

where $\mathfrak{N} = \langle \Phi_0 | \Phi_1 \rangle$ *is a normalization constant and* Z *a skew symmetric matrix. Thouless has given this theorem for pure Slater determinants* Φ_0, Φ_1. *Two-quasi-particle states in this case are ph states*:

$$|\Phi_1\rangle = \mathfrak{N} \cdot \exp\left(\sum_{mi} Z_{mi} a_m^+ a_i \right) |\Phi_0\rangle. \tag{E.27}$$

To prove this theorem we start with two sets of quasi-particle operators β, β^+ and γ, γ^+ belonging to the function $|\Phi_0\rangle$ and $|\Phi_1\rangle$:

$$\begin{aligned} \beta_k^+ &= \sum_l U_{0lk} c_l^+ + V_{0lk} c_l, \\ \gamma_k^+ &= \sum_l U_{1lk} c_l^+ + V_{1lk} c_l, \end{aligned} \tag{E.28}$$

and express the operators γ^+ together with (7.6) by the operators β, β^+:

$$\gamma_k^+ = \sum_{k'} U_{k'k} \beta_{k'}^+ + V_{k'k} \beta_k, \tag{E.29}$$

with

$$\begin{aligned} U &= U_0^+ U_1 + V_0^+ V_1, \\ V &= V_0^T U_1 + U_0^T V_1. \end{aligned} \tag{E.30}$$

As we shall see in Section E.4, the overlap $|\langle \Phi_1 | \Phi_0 \rangle|^2$ is given by $|\det(U)|$. Nonorthogonality of $|\Phi_1\rangle$ and $|\Phi_0\rangle$ therefore means that we can invert U and define the operators

$$\tilde{\gamma}_k^+ = \sum_{k'} U_{k'k}^{-1} \gamma_{k'}^+ = \beta_k^+ + \sum_{k'} Z_{k'k}^* \beta_{k'} \tag{E.31}$$

with the skew symmetric matrix Z [see Eq. (7.5)]:

$$Z := (V U^{-1})^* = - U^{+^{-1}} V^+ = - Z^T. \tag{E.32}$$

Since (E.31) is only a transformation among the creation operators γ^+, $|\Phi_1\rangle$ is also vacuum to the operators $\tilde{\gamma}$. This condition determines $|\Phi_1\rangle$ up to a normalization constant. We therefore have only to show that $\tilde{\gamma}_k$ annihilates the r.h.s. of Eq. (E.26):

$$\tilde{\gamma}_k \exp\left(\sum_{k<k'} Z_{kk'} \beta_k^+ \beta_{k'}^+ \right) |\Phi_0\rangle = e^{\hat{Z}} \left\{ e^{-\hat{Z}} \beta_k e^{\hat{Z}} + \sum_{k'} Z_{k'k} \beta_{k'}^+ \right\} |\Phi_0\rangle \tag{E.33}$$

with

$$\hat{Z} = \sum_{k<k'} Z_{kk'} \beta_k^+ \beta_{k'}^+ . \tag{E.34}$$

Using

$$e^{-\hat{Z}} \beta_k e^{\hat{Z}} = \beta_k - [\hat{Z}, \beta_k] = \beta_k - \sum_{k'} Z_{k'k} \beta_{k'}^+ , \tag{E.35}$$

we see that the r.h.s. of Eq. (E.33) vanishes. This completes the proof of Thouless' theorem and shows that Z is uniquely defined by $|\Phi_1\rangle$. In particular, it does not depend on a C-transformation among the quasi-particles γ^+.

We might ask whether we can represent wave functions $|\tilde{\Phi}_1\rangle$ which are orthogonal to $|\Phi_0\rangle$ in a similar way. The answer is a *generalization of Thouless' theorem*, which states that a wave function $|\tilde{\Phi}_1\rangle$ which is orthogonal to $|\Phi_0\rangle$ can be represented as a multi-quasi-particle state on a function $|\Phi_1\rangle$ which is not orthogonal to $|\Phi_0\rangle$,

$$|\tilde{\Phi}_1\rangle = \tilde{\beta}_{b_1}^+ \cdots \tilde{\beta}_{b_n}^+ |\Phi_1\rangle, \tag{E.36}$$

with quasi-particle operators $\tilde{\beta}_{b_\nu}$ that annihilate $|\Phi_0\rangle$.

To prove this generalization and to give an explicit form for the function $|\Phi_1\rangle$, we use the fact that the transformation (E.29) from $(\beta, \beta^+) \to (\gamma, \gamma^+)$ is a general Bogoliubov transformation. We can therefore apply the Bloch–Messiah theorem again and find that it can be decomposed into a transformation among the operators β^+:

$$\tilde{\beta}_k^+ = \sum_{k'} D_{k'k} \beta_{k'}^+ , \tag{E.37}$$

a special Bogoliubov transformation [(7.12) and (7.13)] to "quasi-particle" operators $\tilde{\gamma}, \tilde{\gamma}^+$ which have the vacuum $|\Phi_1\rangle$ and a C-transformation $\tilde{\gamma}^+ \to \gamma^+$.

In analogy to Eq. (7.18), $|\tilde{\Phi}_1\rangle$ can therefore be written as

$$|\tilde{\Phi}_1\rangle = \prod_{i=1}^{n} \tilde{\beta}_{b_i}^+ \cdot \prod_{p>0} (u_p + v_p \tilde{\beta}_p^+ \tilde{\beta}_{\bar{p}}^+) |\Phi_0\rangle, \tag{E.38}$$

where $b_1 \ldots b_n$ are the blocked levels and $u_p > 0$ does not vanish by definition. We therefore get for $|\Phi_1\rangle$,

$$|\Phi_1\rangle = \prod_{p>0} (u_p + v_p \tilde{\beta}_p^+ \tilde{\beta}_{\bar{p}}^+) |\Phi_0\rangle = \left(\prod_{p>0} u_p \right) \exp\left(\sum_{p>0} \frac{v_p}{u_p} \tilde{\beta}_p^+ \tilde{\beta}_{\bar{p}}^+ \right) |\Phi_0\rangle \tag{E.39}$$

which finishes the proof of (E.36).

The transformation $e^{\hat{Z}} |\Phi_0\rangle$ is not unitary, because it changes the norm of the state. We might ask whether there is also a unitary transformation [RS 77b]

$$|\Phi_1\rangle = e^{i\hat{T}} |\Phi_0\rangle \tag{E.40}$$

with a Hermitian single-particle operator $\hat{T}$ which accomplishes the same. The answer is yes. This can even be accomplished for cases where $|\Phi_1\rangle$ is orthogonal to $|\Phi_0\rangle$.

To find one possible choice, we again start with the transformation (E.27) and decompose the matrices U and V according to the Bloch–Messiah theorem [Eq. (7.8)]. Since $|\Phi_1\rangle$ does not depend on the special choice of the matrix C, we use $C = D^+$ and get

$$U = D\bar{U}D^+, \qquad V = D^*\bar{V}D^+. \tag{E.41}$$

$\bar{U}$ is diagonal with the diagonal elements u_k, and $\bar{V}$ can be written as the product of a diagonal matrix $\tilde{V}$ with the elements v_k and a matrix $\bar{S}$, which has only diagonal elements 0 and 1 and 2×2 blocks of the form $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ along the diagonal line [see (7.9)]

$$\bar{V} = \tilde{V}\cdot\bar{S}. \tag{E.42}$$

If we now define a diagonal matrix $\bar{R}$ with the elements $0 \leqslant r_k < \pi/2$ and $u_k = \cos r_k$, $v_k = \sin r_k$, we get

$$U = \cos R, \qquad V = S\cdot\sin R = (\sin R)^*\cdot S, \tag{E.43}$$

with

$$R = D\bar{R}D^+ \qquad \text{and} \qquad S = D^*\bar{S}D^+ := e^{-iF}, \tag{E.44}$$

where S is unitary and R and F are Hermitian.

We can now define a Hermitian operator $\hat{T}$,

$$\hat{T} = \sum_{k<k'} T_{kk'}\beta_k^+\beta_{k'}^+ + \text{h.c.}, \tag{E.45}$$

which has only a two-quasi-particle part and where the matrix T is given by its "polar decomposition" [Ga 72] as

$$T = iRe^{iF} = ie^{iF}\cdot R^*. \tag{E.46}$$

With the formula $e^A B e^{-A} = B + [A, B] + (1/2!)[A, [A, B]] + \cdots$, it is easy to show that the transformation of the quasi-particles

$$e^{i\hat{T}}\beta_k^+ e^{-i\hat{T}} = \sum_{k'} U_{k'k}\beta_{k'}^+ + V_{k'k}\beta_{k'} \tag{E.47}$$

has the matrices U and V in Eq. (E.43). Equation (E.32) gives a connection to the matrix Z of Thouless' theorem

$$Z = -(\operatorname{tg} R)\cdot e^{iF}, \tag{E.48}$$

which shows that Z diverges if R has eigenvalues $r_k = \pi/2$, that is, $u_k = 0$. The advantage of the transformation $e^{\hat{Z}}$ is that it is uniquely defined by the wave functions $|\Phi_1\rangle$ and $|\Phi_0\rangle$. The matrix T is only unique if we restrict ourselves to eigenvalues r_k of R in the interval $0 \leqslant r_k < \pi/2$.

Those considerations are a special case of a general formula given by Balian and Brézin [BB 69, HI 79], which holds for any Hermitian single-

particle operator $\hat{S}$ bilinear in the operators β_k, β_k^+

$$\hat{S}=\sum_{kk'} S_{kk'}^{11}\beta_k^+\beta_{k'}+\sum_{k<k'}(S_{kk'}^{20}\beta_k^+\beta_{k'}^+ +\text{h.c.}).$$

The unitary operator $e^{i\hat{S}}$ can be written as

$$e^{i\hat{S}}=\langle e^{i\hat{S}}\rangle\cdot e^{\hat{Z}}\,e^{i\hat{Y}}\,e^{\hat{X}},$$

with $\beta|\rangle=0$ and

$$\hat{X}=\sum_{k<k'} X_{kk'}\beta_{k'}\beta_k, \qquad \hat{Y}=\sum_{kk'} Y_{kk'}\beta_k^+\beta_{k'}, \qquad \hat{Z}=\sum_{k<k'} Z_{kk'}\beta_k^+\beta_{k'}^+,$$

The matrices X, Y, and Z can be derived from the matrices U and V obtained from the unitary transformation of the operators β_k^+ [in analogy to Eq. (E.47)]:

$$e^{i\hat{S}}\beta_k^+ e^{-i\hat{S}}=\sum_{k'} U_{k'k}\beta_{k'}^+ + V_{k'k}\beta_{k'},$$

namely:

$$X=V^T U^{+-1}; \qquad Z=V^* U^{*-1}; \qquad e^{iY}=(U^+)^{-1}.$$

The matrices U and V can be expressed by the matrices S^{11} and S^{20} in the following way.

$$\begin{pmatrix} U & V^* \\ V & U^* \end{pmatrix}=\exp\left\{ i\begin{pmatrix} S^{11} & S^{20} \\ -S^{20*} & -S^{11*} \end{pmatrix}\right\}.$$

E.4 The Onishi Formula

In the context of many problems (for instance, in the GCM method), we need the overlap integrals $\langle\Phi_1|\Phi_0\rangle$ of the norm and of other operators $\langle\Phi_1|\hat{O}|\Phi_0\rangle$ for general product wave functions $|\Phi_0\rangle$ and $|\Phi_1\rangle$. In the case of pure Slater determinants such formulas have been given by Löwdin [Lö 55]. They have been generalized for HFB wave functions by Onishi and Yoshida [OY 66]. A very elegant derivation, which can also be applied to the boson case, has been given by Balian and Brézin [BB 69].

These formulas can only be used if the overlap $\langle\Phi_1|\Phi_0\rangle$ does not vanish. In cases where it vanishes we can use the representation (E.36) for $|\Phi_1\rangle$.

With the definitions (E.30), (E.32), and (E.43) the overlap integral for the norm is given by*

$$\langle\Phi_1|\Phi_0\rangle=\sqrt{\det U}=\sqrt{\det(U_0^+U_1+V_0^+V_1)}=\sqrt{\det(\cos R)}\,. \quad \text{(E.49)}$$

Before we give the proof of this formula we rewrite it in different ways. In the basis where U is diagonal, it is easy to show that

$$\det(U)=\exp(\mathrm{Tr}\ln U). \quad \text{(E.50)}$$

*Since the wave functions $|\Phi_0\rangle$ and $|\Phi_1\rangle$ from the definition (7.15) are only given to a phase, there is also a phase open in the following formulas. We assume U to be Hermitian and positive definite (see Eq. E.6).

Using the transformations (E.28), we express the wave functions $|\Phi_i\rangle$ $(i=0,1)$ by the operators c_l^+ and the bare vacuum $|-\rangle$ (which is only possible if $\langle\Phi_i|-\rangle\neq 0$), and get from (E.49) with $Z_i=(V_iU_i^{-1})^*$

$$\langle -|e^{\hat{Z}_1^+}e^{\hat{Z}_0}|-\rangle=\exp\{\tfrac{1}{2}\mathrm{Tr}(\ln(1-Z_1^*Z_0))\}. \tag{E.51}$$

Next we give the formula for the Hamilton overlap:

$$\langle\Phi_1|H|\Phi_0\rangle=\langle\Phi_1|\Phi_0\rangle\cdot\{\mathrm{Tr}(\epsilon\rho^{10})+\tfrac{1}{2}\mathrm{Tr}_1\mathrm{Tr}_1(\rho^{10}\bar{v}\rho^{10})+\tfrac{1}{4}\mathrm{Tr}_2\mathrm{Tr}_2(\kappa^{01*}\bar{v}\kappa^{10})\}. \tag{E.52}$$

Where the notation of Eq. (E.19) is used, the transition densities ρ^{10},κ^{10} and κ^{01} are defined by

$$\rho_{ll'}^{10}=\frac{\langle\Phi_1|c_{l'}^+c_l|\Phi_0\rangle}{\langle\Phi_1|\Phi_0\rangle}, \tag{E.53a}$$

$$\kappa_{ll'}^{10}=\frac{\langle\Phi_1|c_{l'}c_l|\Phi_0\rangle}{\langle\Phi_1|\Phi_0\rangle}, \tag{E.53b}$$

$$\kappa_{ll'}^{01*}=\frac{\langle\Phi_1|c_l^+c_{l'}^+|\Phi_0\rangle}{\langle\Phi_1|\Phi_0\rangle}. \tag{E.53c}$$

These matrices can be expressed by the densities ρ_0,κ_0, the HFB coefficients U_0,V_0, which belong to $|\Phi_0\rangle$, and by the Thouless matrix Z [Eq. (E.32)], or after some calculations with the orthogonality relations for U_0,V_0,U_1,V_1 in a more symmetric way by Z_0 and Z_1 [see Eq. (E.51)]:

$$\rho^{10}=\rho_0-V_0^*Z^*U_0^+=-Z_0(1-Z_1^*Z_0)^{-1}Z_1^* \quad =V_0^*U^{T^{-1}}V_1^T, \tag{E.54a}$$

$$\kappa^{10}=\kappa_0-V_0^*Z^*V_0^+= \quad Z_0(1-Z_1^*Z_0)^{-1} \quad =V_0^*U^{T^{-1}}U_1^T, \tag{E.54b}$$

$$\kappa^{01*}=\kappa_0^*+U_0^*Z^*U_0^+= \quad (1-Z_1^*Z_0)^{-1}\ Z_1^*=-U_0^*U^{T^{-1}}V_1^T. \tag{E.54c}$$

We see that for $|\Phi_1\rangle\to|\Phi_0\rangle$, these densities go over into the usual densities ρ,κ,κ^*. In the same way we can calculate the overlap integral for arbitrary operators expressed by c_l^+ and c_l. We must first derive the expectation value with respect to a HFB wave function. According to Wick's theorem (C.4), we can express such expectation values by the contractions $\rho\sim\overbrace{c^+c}$, $\kappa\sim\overbrace{cc}$; $\kappa^*\sim\overbrace{c^+c^+}$. To get the overlap integral, we only have to replace ρ by ρ^{10}, κ by κ^{10}, and κ^* by κ^{01*}, and multiply by $\langle\Phi_1|\Phi_0\rangle$. Obviously, these conclusions are also valid if we replace the bare vacuum by an arbitrary HFB function $|\Phi\rangle$ and the operators c^+,c by the corresponding quasiparticle operators.

In the following we present a derivation of the formulae (E.49) and (E.54):

To get $\langle\Phi_1|\Phi_0\rangle$ we present $|\Phi_1\rangle$ in the canonical basis of the transformation (E.29). This is done in Eq. (E.38). Since $\langle\Phi_1|\Phi_0\rangle$ does not vanish, there are no blocked levels and we get $\langle\Phi_1|\Phi_0\rangle$ as a product of all the real positive numbers u_p. These are the two-fold degenerate eigenvalues of the

matrix U [Eq. (E.41)]:

$$\langle\Phi_1|\Phi_0\rangle = \prod_{p>0}(u_p) = \Big(\prod_{p\gtrless 0} u_p\Big)^{1/2} = (\det U)^{1/2}. \tag{E.55}$$

Next we derive the overlap integral for an arbitrary operator expressed by the basis operators c, c^+. From Thouless's theorem, with (E.34), we get

$$|\Phi_1\rangle = e^{\hat{Z}}|\Phi_0\rangle\langle\Phi_0|\Phi_1\rangle, \qquad e^{-\hat{Z}^+}|\Phi_0\rangle = |\Phi_0\rangle, \tag{E.56}$$

and

$$\langle\Phi_1|c_{l_1}^+\cdots c_{l_p}^+ c_{k_1}\cdots c_{k_q}|\Phi_0\rangle = \langle\Phi_1|\Phi_0\rangle\langle\Phi_0|\bar{d}_{l_1}\cdots\bar{d}_{l_p}d_{k_1}\cdots d_{k_q}|\Phi_0\rangle, \tag{E.57}$$

where we have introduced the operators

$$\bar{d}_l = e^{\hat{Z}^+}c_l^+ e^{-\hat{Z}^+} \quad \text{and} \quad d_l = e^{\hat{Z}^+}c_l e^{-\hat{Z}^+}. \tag{E.58}$$

These operators obey Fermi commutation relations. We can therefore apply Wick's theorem (C.4) to evaluate the matrix element (E.53), which shows that we need only the contractions $\bar{d}\bar{d}$, $\widehat{dd}$, and $\widehat{\bar{d}\bar{d}}$. They are given by ρ^{10}, κ^{10}, and κ^{01*}, respectively, in Eqs. (E.53) and can be calculated by expressing the operators $d, \bar{d}$ in terms of the operators β, β^+:

$$\begin{aligned}\bar{d}_l &= \sum_k U_{0lk}^*\beta_k^+ + \sum_k (V_0 + U_0^* Z^*)_{lk}\beta_k,\\ d_l &= \sum_k (U_0 + V_0^* Z^*)_{lk}\beta_k + \sum_k V_{0lk}^*\beta_k^+.\end{aligned} \tag{E.59}$$

E.5 Bogoliubov Transformations for Bosons

As in the case of fermions, starting from a basis set of boson operators B_μ, B_μ^+ with the "bare" vacuum $|-\rangle$, we can make a Bogoliubov transformation to other bosons O_ν, O_ν^+ as

$$\begin{pmatrix} B \\ B^+ \end{pmatrix} = \begin{pmatrix} X & Y^* \\ Y & X^* \end{pmatrix}\begin{pmatrix} O \\ O^+ \end{pmatrix} = \mathfrak{X}\begin{pmatrix} O \\ O^+ \end{pmatrix}, \tag{E.60}$$

in which the matrix $\mathfrak{X}$ is unitary with respect to the metric

$$\mathfrak{N} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \mathfrak{X}\mathfrak{N}\mathfrak{X}^+ = \mathfrak{N}, \qquad \mathfrak{X}^+\mathfrak{N}\mathfrak{X} = \mathfrak{N}. \tag{E.61}$$

Corresponding to the Bloch–Messiah theorem (E.1), we can show that each matrix $\mathfrak{X}$ of this kind can be decomposed in the following way.

$$\mathfrak{X} = \begin{pmatrix} D & 0 \\ 0 & D^* \end{pmatrix}\begin{pmatrix} \overline{X} & \overline{Y} \\ \overline{Y} & \overline{X} \end{pmatrix}\begin{pmatrix} C & 0 \\ 0 & C^* \end{pmatrix}, \tag{E.62}$$

where D and C are unitary and $\overline{X}$ and $\overline{Y}$ are diagonal with real nonnegative diagonal elements x_k, y_k with the relation $x_k^2 - y_k^2 = 1$ (Bloch–Messiah theorem for bosons).

An operator *quadratic in bosons*

$$\hat{F}=\sum_{\mu\nu} f_{\mu\nu}B_\mu^+ B_\nu+\tfrac{1}{2}\sum_{\mu\nu}(g_{\mu\nu}B_\mu^+ B_\nu^+ +\text{h.c.}) \tag{E.63}$$

may be expressed by the new bosons O, O^+ in the following way.

$$\hat{F}=\tfrac{1}{2}\mathrm{Tr}(F^{11}-f)+\sum_{\mu\nu}F^{11}_{\mu\nu}O_\mu^+ O_\nu+\tfrac{1}{2}\sum_{\mu\nu}(F^{20}_{\mu\nu}O_\mu^+ O_\nu^+ +\text{h.c.}), \tag{E.64}$$

with

$$F^{11}=X^+fX\ +Y^+f^*Y\ +X^+gY\ +Y^+g^*X, \tag{E.65}$$

$$F^{20}=X^+fY^*+Y^+f^*X^*+X^+gX^*+Y^+g^*Y^*. \tag{E.66}$$

The vacuum $|\Phi)$ belonging to the bosons O is uniquely defined by

$$O_\mu|\Phi)=0 \qquad \text{for all } \mu. \tag{E.67}$$

If we have two such vacua $|\Phi_0)$ and $|\Phi_1)$ with the operators $O^{(0)}$ and $O^{(1)}$, and if we define

$$X=X_0^+X_1-Y_0^+Y_1; \qquad Y=X_0^TY_1-Y_0^TX_1, \tag{E.68}$$

we can prove a *Thouless' Theorem for bosons**:

$$|\Phi_1)=(\Phi_0|\Phi_1)\cdot\exp\Big(\tfrac{1}{2}\sum_{\mu\nu}Z_{\mu\nu}O_\mu^{(0)+}O_\nu^{(0)+}\Big)|\Phi_0) \tag{E.69}$$

with

$$Z=(YX^{-1})^*=(X^+)^{-1}Y^+=Z^T. \tag{E.70}$$

This representation is always possible, because vacua that are connected by a finite Bogoliubov transformation are never orthogonal. In contrast to the fermion case, it is impossible to represent a multi-boson state $O_1^+\ldots O_n^+|\Phi_0)$ as a new vacuum for new bosons. In analogy to (E.40), we can also write

$$|\Phi_1)=e^{i\hat{T}}|\Phi_0), \tag{E.71}$$

where

$$\hat{T}=\tfrac{1}{2}\sum_{\mu\nu}T_{\mu\nu}O_\mu^{(0)+}O_\nu^{(0)+}+\text{h.c.} \tag{E.72}$$

and the matrix T can be found in analogy Eq. (E.46) as

$$T=-iD\bar{R}D^T, \tag{E.73}$$

where the matrix $\bar{R}$ is a diagonal matrix with $\sinh r_k=y_k$ and D and y_k are given by the decomposition (E.62) of the transformation (E.68). With $R=D\bar{R}D^+$, and $S=D^*D^+=e^{-iF}$ we obtain

$$T=-iRe^{iF}, \qquad Z=\mathrm{tgh}(R)e^{iF}. \tag{E.74}$$

* In the boson case simple shift operations are also possible. They give a linear term in the exponent of (E.69). In order not to complicate the formulae, we treat them later [Eq. (E.77)].

For the overlap integral $(\Phi_1|\Phi_0)$ we get, in analogy to the Onishi formulas (E.49ff.),

$$(\Phi_1|\Phi_0)=(\det X)^{-1/2}=(\det(X_0^+X_1-Y_0^+Y_1))^{-1/2}=\exp(-\tfrac{1}{2}\mathrm{Tr}\ln X). \tag{E.75}$$

Overlap integrals of general operators are calculated again by using Wick's theorem and the transition densities

$$\begin{aligned}
\rho^{10}_{\mu\mu'}&=\frac{(\Phi_1|B_{\mu'}^+B_\mu|\Phi_0)}{(\Phi_1|\Phi_0)}=(\rho_0+Y_0^*Z^*X_0^+)_{\mu\mu'} &&=Z_0(1-Z_1^*Z_0)^{-1}Z_1^*,\\
\kappa^{10}_{\mu\mu'}&=\frac{(\Phi_1|B_{\mu'}B_\mu|\Phi_0)}{(\Phi_1|\Phi_0)}=(\kappa_0+Y_0^*Z^*Y_0^+)_{\mu\mu'} &&=Z_0(1-Z_1^*Z_0)^{-1}, \qquad \text{(E.76)}\\
\kappa^{01*}_{\mu\mu'}&=\frac{(\Phi_1|B_\mu^+B_{\mu'}^+|\Phi_0)}{(\Phi_1|\Phi_0)}=(\kappa_0^*+X_0^*Z^*X_0^+)_{\mu\mu'}= &&(1-Z_1^*Z_0)^{-1}Z_1^*,
\end{aligned}$$

where

$$\rho_0=Y_0^*Y_0^T; \qquad \kappa_0=Y_0^*X_0^T; \qquad Z_0=Y_0^*X_0^{*-1}; \qquad Z_1=Y_1^*X_1^{*-1}.$$

In the boson case there exists still a further type of linear transformation to a new set of boson operators, namely *translations* or shift operations

$$O_\mu^+=B_\mu^++c_\mu^* \tag{E.77}$$

with complex numbers c_μ. The vacuum, which belongs to O_μ, has the form of a *coherent state*

$$|\Phi_0)=\mathfrak{N}\exp\Big(-\sum_\mu c_\mu B_\mu^+\Big)|-), \tag{E.78}$$

and the overlap between two boson vacua is

$$(\Phi_1|\Phi_0)=\exp\Big(-\tfrac{1}{2}\sum_\mu|c_\mu^1-c_\mu^0|^2\Big). \tag{E.79}$$

In such cases there also exist non-vanishing expectation values of single bosons:

$$(\Phi_1|B_\mu|\Phi_0)=-(\Phi_1|\Phi_0)c_\mu^0. \tag{E.80}$$